



Unanimity versus majority: proposing under incomplete information

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Abstract

We examine the trade-offs between unanimity and majority voting under incomplete information. While conventional wisdom suggests that unanimity leads to higher reform rejection rates and higher total transfers compared to the majority rule, we show that this does not always hold when the proposer lacks perfect knowledge of responders' acceptance costs. Using a simple theoretical model, we identify conditions under which uncertainty leads the proposer to adopt different bargaining behaviors, sometimes resulting in a higher probability of reform approval under unanimity rule, and sometimes yielding higher total transfers under majority rule. Our findings inform the current EU debate on replacing unanimity with majority voting and provide nuanced insights into the arguments advanced by each side.

Keywords Unanimity · Majority · Bargaining · Voting · Incomplete information

JEL classification D71 · D72

1 Introduction

Decision-making in the Council of the European Union operates at the intersection of economic rationality, political bargaining, and institutional constraints. While qualified majority voting is the default, a set of sensitive dossiers—such as EU enlargement, social security, taxation—require unanimity. In these unanimity-constrained settings, countries with great

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influence (henceforth, *the proposer*) usually initiate discussions on significant reforms with other member states (henceforth, *the responders*). The final proposal that reaches formal approval is the outcome of a series of bilateral and multilateral bargaining rounds between the proposer and the responders, during which the latter may push for revisions, adjustments in other policy areas, and even compensatory transfers.

Unanimity, despite occasionally leading to veto threats or even blackmail – practices seemingly at odds with the EU’s declared cooperative principles – has long been justified on economic grounds.¹ By requiring the approval of all member states, unanimity ensures that no country is made worse off. In particular, if every policy reform is presented for a vote alongside a strategically designed redistribution scheme, unanimity rule guarantees the adoption of an efficient reform with appropriately structured transfers, leading to a Pareto improvement over the status quo (Wicksell 1896, 1958; Musgrave 1939; Buchanan and Tullock 1962; Buchanan 1975).² Moreover, in some high-stakes cases unanimity has arguably operated as a constructive constraint rather than as a means for extortion. A well-known example is Denmark’s rejection of the Maastricht Treaty in its 1992 referendum, which – because unanimity was required – temporarily blocked ratification for all member states. Rather than bypass the Danish electorate, as might have been possible under qualified majority voting, and thereby call into question Denmark’s participation in the EU, the European Council negotiated the Edinburgh Agreement, granting Denmark opt-outs in areas such as the monetary union, defense, and home affairs. With these changes in place, Danish voters approved the Maastricht package in a second referendum in 1993, preserving both the depth of integration desired by most states and Denmark’s continued participation (Svensson 1994; Howarth 1994).³

While unanimity protects national interests and promotes cooperation, it also fosters bargaining inefficiencies. Since every decision requires extensive bilateral and multilateral bargaining under unanimity, responders can leverage their veto power to extract concessions, policy adjustments, or compensatory transfers, sometimes at the expense of timely and effective decision-making.⁴ This strategic behavior has led to prolonged negotiations,

¹ Critics argue that veto-related practices prioritize short-term geopolitical interests over core EU values, including the rule of law and human dignity; see, for instance, the European Youth Portal article “Is the Veto Holding the EU Back?” https://youth.europa.eu/news/veto-holding-eu-back_en. Academic observers additionally warn of “vetocracy,” where strategic vetoes may conflict with the rule of law and predictable governance (Bellenghi 2022). However, the unanimity rule secures Pareto-efficient outcomes by creating a “unanimity core” that no single member can overturn (Tsebelis 2002), fosters credible commitment and investment stability by protecting property-rights expectations (Moravcsik 1998), allows compensatory side-payments to internalize redistributive externalities (Bailer et al. 2015; Mattila and Lane 2001), and deters free-riding by aligning all members’ incentives *ex ante* (Cecchini 1988).

² For a detailed comparison of Buchanan’s and Musgrave’s readings of Wicksell, see Johnson (2006). For a general survey of unanimity and related voting rules, see Mueller (2003). From an economic perspective, this structured process prevents detrimental reforms by allowing for tailored compensation mechanisms that align incentives across member states.

³ In principle, unanimity equalizes the power of all states and is therefore fair from a one-state-one-vote perspective. Qualified majority systems, by contrast, assign larger numbers of votes to larger countries (for example, under the EU’s qualified majority system Germany, the largest member state, holds nearly 20% of all votes), aspiring instead to be fair from a one-person-one-vote perspective. However, as power-index analyses typically show, such systems magnify the voting power of large players—even after accounting for their larger size—thus giving them disproportionate advantages (Leech 1988; Laruelle and Widgrén 1998).

⁴ Levy (2017) highlights that supranational institutions such as the EU emerge precisely to reduce the transaction and enforcement costs associated with collective action, thereby allowing for a more efficient realization of shared policy objectives. McGann (2004) argues that, while super-majority decision rules are often

blocked reforms, and, in some cases, outright policy paralysis. An example is Hungary's veto threats and resistance to EU-wide sanctions against Russia,⁵ illustrating how unanimity can be exploited in bargaining to delay collective action. Majority rule, by contrast, reduces the likelihood of gridlock by allowing decisions to pass even when some members dissent.

Calls for reform have led to a shift toward (super-)majoritarian decision-making in various policy domains, but many key areas of EU policymaking still require unanimity. The report of Mario Draghi on European competitiveness (Draghi 2024) argues that to ensure the EU remains globally competitive, institutional reforms must accelerate decision-making, particularly by extending qualified majority voting (QMV) in the Council of the EU.⁶ In fact, this transition has already started, as in several domains of policy in the EU, a special kind of super-majority rule is applied. However, as far as key decisions are concerned the consensus procedure is still in place, because smaller states fear that transitioning entirely to majoritarian rules would weaken their bargaining power, reducing their policy influence and access to transfers.⁷

The established view in the literature and policy debate is that unanimity requirements in collective decision-making, such as those used in key areas of EU policymaking, tend to increase both the likelihood of reform rejection and the total transfers required to secure agreement, compared to majority voting. Under unanimity, every member has veto power, enabling even a single state to block reforms unless sufficiently compensated, which often leads to protracted bargaining, costly side payments, and sometimes outright policy paralysis. In contrast, majority rule is believed to streamline decision-making, reduce the threat of gridlock, and allow proposers to secure reform with lower total transfers by focusing offers on a minimal winning coalition, thus making reform both more likely and less expensive overall.

Our analysis challenges this conventional wisdom by showing that, under incomplete information – when proposers do not know the exact acceptance costs of other members – these presumed advantages of majority rule do not always hold. Using a simple model, we demonstrate that uncertainty can induce the proposer to adopt more cautious strategies under unanimity (offering sufficiently high transfers to all) and riskier, less generous strategies under majority rule (offering small transfers in the hope that at least a sufficiently large number of responders will accept). As a result, there are plausible scenarios in which the probability of reform acceptance is actually higher under unanimity, and the total transfers required for acceptance are lower than under majority rule. These findings overturn standard intuitions and highlight that the relative efficiency of voting rules depends crucially on the level of information the proposer has about responders' preferences.

Our theoretical framework, although highly stylized, closely mirrors the realities of EU decision-making, where large member states typically act as proposers and must navigate significant uncertainty about the political preferences and acceptance costs of other member states. In practice, this “incomplete information” reflects the difficulty proposers face

justified in terms of minority protection, they can also create decision-making paralysis and privilege the status quo, preventing necessary reforms even when they are broadly beneficial.

⁵ See, for instance, <https://www.ft.com/content/6ea5f0eb-15ed-4958-9ce0-f9cd93556de5>

⁶ The report highlights how unanimity requirements have hindered deeper integration efforts and delayed necessary economic and technological reforms. It recommends utilizing the passerelle clause—a mechanism allowing a shift from unanimity to QMV without treaty change to overcome these roadblocks.

⁷ See, for instance, Schuette (2019).

in gauging the domestic constraints and true willingness of their counterparts to accept reforms, especially in sensitive policy areas. As in our model, large states strategically design offers – often in the form of policy concessions – without precise knowledge of what will secure consent. The trade-offs we identify are not merely theoretical curiosities, but capture the core strategic dilemmas faced in EU institutional bargaining, where the choice of voting rule interacts with uncertainty to shape both the likelihood of reform and the distribution of transfers.

While some results may seem to contradict conventional wisdom, they are grounded in non-trivial yet solid economic intuition. When there is sufficient uncertainty regarding the responders' acceptance costs, a proposer under majority rule may gamble by offering each responder a small transfer, knowing that the probability that a sufficient number of them will accept is substantial. Under unanimity, however, the proposer instead adopts a safer strategy and offers larger transfers to all responders, because the probability that everyone would accept a low offer is too small. This difference between insurance and gambling can lead to a higher probability of reform approval under unanimity than under majority rule. Moreover, when responders are likely to accept only very high offers, a proposer under majority rule may choose to make such a high offer to a subset of responders in order to secure the required number of votes. Under unanimity, by contrast, the proposer may optimally risk smaller offers to all responders, since making very high offers to everyone would be prohibitively expensive. In some instances, the total cost of these smaller offers under unanimity can be lower than the amount needed to secure a sufficient number of votes under majority rule. Thus, there are situations in which reforms pass under both unanimity and majority rule, but total transfers are higher under the latter.

Our results do not contradict the argument that proposers hold a significant advantage under the majority rule and that responders' interests are generally better served by unanimity. Rather, we qualify this view by emphasizing that it should not be regarded as an unambiguous truth that applies in all cases. In some scenarios, proposers can benefit under unanimity, and responders might gain from a transition to a majoritarian system. This observation helps depolarize the ongoing debate, mitigating both the urgency of reform and the resistance to it. We believe this paper contributes to this discussion by demonstrating, through a solid rational-choice equilibrium approach, that both proposers and responders might experience gains or losses from relaxing the unanimity requirement in decision-making processes.

2 Related literature

This paper contributes to the extensive literature on multilateral legislative bargaining particularly the branch that investigates how different voting rules affect bargaining outcomes. The canonical reference point here is the multi-round, complete-information model of Baron and Ferejohn (1989), where an agenda setter can repeatedly reallocate a fixed surplus until a proposal passes or the legislature dissolves. Because all pay-offs are common knowledge, equilibrium agreement is immediate—so welfare depends only on the voting threshold: simple-majority coalitions are minimal and exclude costly members, whereas unanimity spreads the surplus more evenly but can inflate the agenda setter's price of passage.

Recent experimental evidence underscores the empirical relevance of these institutional differences. Agranov et al. (2024) implement a stochastic-budget design and show that when the available surplus evolves randomly, unanimity can outperform majority rule by encouraging delay only when the expected future gains justify it. Their findings challenge the traditional view that unanimity necessarily produces excessive delay and inefficiency, illustrating how uncertainty—whether about future resources or about agents’ preferences—can reverse standard welfare rankings. Our model provides a complementary theoretical rationale for such reversals by showing that incomplete information about acceptance costs can similarly make unanimity more effective than majority rule.

Our framework retains the same three-player setting as in Baron and Ferejohn (1989) (one proposer, two responders) but collapses the horizon to a single proposal, introducing private information about each responder’s acceptance cost. Conceptually, this one-shot structure can be viewed as a limiting case of dynamic models such as those in Tsai and Yang (2010) and Piazzolo and Vanberg (2025), where bargaining unfolds over multiple periods and rejection serves as a signal about types. By shutting down this dynamic channel, our static setup abstracts from learning and signaling but gains tractability and focus: it isolates the pure informational friction introduced by hidden acceptance costs and allows for a clean comparison of unanimity and majority rules without confounding effects from delay or reputation formation.⁸

The closest paper to ours is the two-period model of Piazzolo and Vanberg (2025), who study both unanimity and simple-majority rule under private information in an otherwise standard Baron-Ferejohn environment. In their setting, a first-round rejection can reveal who is “expensive,” prompting the proposer to revise the offer in a second period. They find that unanimity typically decreases the probability of successful policy implementation compared to majority voting due to increased bargaining frictions and heightened veto threats under incomplete information. In contrast, we provide conditions under which unanimity can paradoxically enhance the likelihood of successful policy passage. The same rejection-as-signal logic also drives the earlier majority-only analysis of Tsai and Yang (2010). By contrast, we shut down the dynamic channel: the proposer has a single chance to allocate either zero, a low transfer, or a high transfer to each responder, and if the offer fails, the status quo prevails. With no second stage, the entire strategic problem reduces to choosing between a risky, cheap package and a safe, costly package. This one-shot design retains the key informational friction but strips away delay and renegotiation, letting us pinpoint the pure effect of the voting threshold on efficiency and transfers.⁹

We also note the recent contribution of Rosenthal and Zame (2022), who analyse sequential agenda-setting when the proposer is uncertain about a voter’s ideal point and confronts a continuum of potential supporters. Although their focus is on spatial policy (with no trans-

⁸The closest analogue to a dynamic model that retains the uncertainty our result requires is to add a preliminary cheap-talk stage in which responders may send cost messages. As we show in Section 7, for the parameter values that generate our paradox, these messages cannot credibly reveal private acceptance costs, the proposer’s uncertainty persists. Hence the logic behind our comparison of unanimity and majority remains unchanged.

⁹Our setting also relates to the reputational bargaining literature, in which some players may be privately committed to a hard stance and rational types can benefit from mimicking them. Ma (2023) extends the Abreu and Gul (2000) framework to a multilateral (Baron and Ferejohn 1989) game and shows that majority rule weakens the incentive to cultivate a “tough” reputation—because costly types risk exclusion from minimal winning coalitions—whereas unanimity preserves it. Although our model contains no reputational types, it shares the core insight that the voting threshold itself shapes the payoff to appearing hard to satisfy.

fers) and the game unfolds over multiple proposals, their core message—that uncertainty about a pivotal voter’s preferences crucially shapes the proposer’s strategy—parallels our finding that hidden acceptance costs change the optimal offer.

Pure one-shot bargaining models are scarce. An exception is Chen and Eraslan (2014), who study a three-person game with one proposer and two responders. Each responder privately observes the intensity of his ideological preferences, which determines both the policy he prefers and the transfer he must receive to vote “yes.” The voting rule is simple majority, so the proposer needs the support of either responder. Before making her single take-it-or-leave-it offer, the proposer listens to cheap-talk messages from the responders. Chen and Eraslan (2014) characterize when such communication credibly reveals types and show that competition between the two responders typically dampens information disclosure, leaving the proposer worse off than in a baseline two-player veto setting (one proposer, one responder) where no competition exists. A companion study, Chen and Eraslan (2013), reaches similar conclusions in a purely spatial model with privately known ideal points instead of ideological intensities.

Our study also contributes to the broader debate over whether unanimity or majority rule yields higher welfare. When side-payments are ruled out, theory offers no universal ranking: with preference or signal uncertainty, unanimity can hinder information aggregation (Feddersen and Pesendorfer 1998; Austen-Smith and Feddersen 2006), whereas with heterogeneous stakes, weighted or veto-augmented majorities can Pareto-dominate both simple majority and unanimity (Azrieli 2018; Bouton et al. 2018). Laboratory evidence is equally context-sensitive. Jury experiments by Guarnaschelli et al. (2000) and the information-aggregation study of Bouton et al. (2017) find majority (or majority-with-veto) superior in some signal structures, while other treatments reverse that ordering. Within the experimental bargaining branch, Miller and Vanberg (2013, 2015) show that with a fixed deterministic budget, unanimity generates longer and socially wasteful delays than majority. By contrast, as already noted, Agranov et al. (2024) show that when the surplus evolves stochastically, unanimity can dominate by fostering efficient patience. Taken together, these studies suggest that unanimity tends to perform better when uncertainty—about future gains or, in our framework, acceptance costs—is sufficiently high, whereas majority prevails when the surplus is fixed and commonly known.¹⁰

Finally, because our model is a one-shot, take-it-or-leave-it proposal followed by a vote, it naturally connects to the ultimatum-bargaining literature—particularly the strand that examines how incomplete information shapes the outcome of a single offer. Some of the most recent papers that, like ours, consider private information in a one-shot environment are those of Gonzalez-Sanchez and Loyola (2024) and Konrad and Thum (2021). In the bilateral domain, Gonzalez-Sanchez and Loyola (2024) let the proposer know her own “envy” parameter while the responder observes only the monetary split. The hidden type forces the proposer to balance a low-probability but equitable offer against a high-probability, unequal one—exactly the risk-return trade-off that drives our model. Moving to a genuinely multi-lateral setting, Konrad and Thum (2021) study an all-at-once ultimatum over global tax coordination and show that a single proposal can achieve full cooperation more easily than sequential negotiations when countries’ valuations are unobserved. Both papers confirm that incomplete information alone can reshape a single-offer bargain.

¹⁰ See Palfrey (2016) and Agranov (2022) for comprehensive surveys of experimental legislative bargaining.

The paper is organized as follows. We first provide a simple numerical exercise that illustrates how the voting rules influence the strategic incentives of the proposer and, thus, lead to the seemingly paradoxical results in our further analysis in Section 3. In Section 4, we introduce our basic model. Section 5 presents the equilibrium analysis for both unanimity and majority rules. In Section 6, we compare these two rules under uncertainty and transfers, which forms our main contribution. In Section 7, we discuss the robustness of our results by relaxing some of the baseline assumptions and considering extensions of the model. Finally, Section 8 concludes the paper.

3 Two simple examples

To illustrate how incomplete information creates different incentives under majority and unanimity voting, we consider a very simple numerical example. This example will also allow us to point out the mechanisms behind one of the seemingly paradoxical results of our paper. Suppose that the proposer and the two responders are negotiating to split a dollar. The proposer proposes a split, and then the responders vote to accept or reject the split, either using the unanimity rule or the majority rule. A responder's fallback option or cost of acceptance – unknown to the proposer – is either 0.05 with probability $\frac{3}{4}$ or 0.2 with probability $\frac{1}{4}$; while the proposer's fallback option is zero. In other words, we consider two responders $i \in \{2, 3\}$ having i.i.d. acceptance costs

$$c_i = \begin{cases} 0.05 & \text{with prob. } \frac{3}{4}, \\ 0.20 & \text{with prob. } \frac{1}{4}, \end{cases}$$

The proposed transfers (t_2, t_3) must satisfy $t_2, t_3 \geq 0$ and $t_2 + t_3 \leq 1$. All players are risk-neutral: utilities are linear in transfers and costs. A responder accepts iff $t_i \geq c_i$ and the proposer maximizes expected payoff, where $1 - (t_2 + t_3)$ is the payoff upon acceptance. Note that the proposer will offer either 0.2 or 0.05 as transfers; offering more than the fallback options does not increase the acceptance probability but reduces the proposer's payoff in case of success.

In a full-information benchmark, the proposer perfectly observes the responders' acceptance costs; the equilibrium proposal is accepted under both voting rules. Under unanimity, the proposer offers each responder her fallback payoff and earns 0.9 (if both are low types), 0.75 (if one is low and one is high), or 0.6 (if both are high). Under majority rule, the proposer only needs to compensate the least expensive responder and can thus earn 0.95 if at least one responder has low costs, or 0.8 if both are high-cost types.

How do things change with incomplete information? Table 1 provides a summary of the expected payoffs for the proposer and the associated acceptance probabilities under both rules. We highlight the equilibrium payoffs and probabilities of success in bold.

With the unanimity rule, the proposer can play the safe strategy and offer 0.2 to both responders. This leads to successful negotiations with certainty and yields the proposer a payoff of 0.6 ($= 1 - .2 - .2$). Alternatively, the proposer can gamble either by offering a low transfer (0.05) to both responders or one low and one high transfer. The first strategy is successful only if both responders are of the low type (probability $\frac{3}{4} \cdot \frac{3}{4} = 0.563$). The second gambling strategy is successful if the responder with the low offer is also of the

Table 1 Proposer's expected payoffs and probabilities of acceptance under unanimity and majority ($\underline{c} = 0.05$, $\bar{c} = 0.20$)

Offers (t_2, t_3)	(p_2, p_3)	Success U	Success M	Transfers	Payoff U	Payoff M
(0.05, 0.05)	(0.75, 0.75)	0.56	0.94	0.100	0.506	0.844
(0.20, 0)	(1.00, 0.00)	0.00	1.00	0.200	0.000	0.800
(0.20, 0.05)	(1.00, 0.75)	0.75	1.00	0.250	0.563	0.750
(0.20, 0.20)	(1.00, 1.00)	1.00	1.00	0.400	0.600	0.600

Table 2 Proposer's success probabilities and payoffs under unanimity and majority ($\underline{c} = 0.20$, $\bar{c} = 0.45$, $p = 0.35$)

Offers (t_2, t_3)	(p_2, p_3)	Success U	Success M	Transfers	Payoff U	Payoff M
(0.20, 0.20)	(0.65, 0.65)	0.42	0.88	0.400	0.254	0.526
(0.45, 0)	(1.00, 1.00)	0.00	1.00	0.450	0.000	0.550
(0.45, 0.20)	(1.00, 0.65)	0.65	1.00	0.650	0.227	0.350
(0.45, 0.45)	(1.00, 1.00)	1.00	1.00	0.900	0.100	0.100

low type, which happens with probability $\frac{3}{4}$. Overall, the two gambling strategies generate expected payoffs of .51 ($= (1 - 2 \cdot .05) \cdot \frac{9}{16}$) and .56 ($= (1 - 0.2 - 0.05) \cdot \frac{3}{4}$). Obviously, the proposer is best off by playing the safe strategy. The proposer behaves in the same way as if she faces two high types in the case of complete information.

With the majority rule, a safe strategy implies offering .2 to one responder (and zero to the other). The responder with the generous offer will accept, regardless of whether he is of the high or low type. The proposer's payoff amounts to 0.8. However, playing a risky strategy is even more profitable. If the proposer offers .05 to both responders, this will reduce the probability of success only a bit, as only one acceptance is needed. The probability that at least one responder is a low type amounts to 0.94 ($= 1 - \frac{1}{4} \cdot \frac{1}{4}$). Hence, the proposer's expected payoff becomes .84 ($= .9 \cdot \frac{15}{16}$).

This example illustrates that under incomplete information, the tradeoffs can be very different for majority and unanimity rules. In our numerical example, deviating from the safe strategy saves transfer payments at the cost of a small reduction in success probability under the majority rule, while there is a significant loss of success probability under unanimity. These differences in the trade-offs lead to the somewhat surprising result that the acceptance of a (welfare improving) reform may be more likely under the unanimity rule than under majority rule.

A similar mechanism is behind our second counterintuitive result that the transfer payments might be higher under the majority rule.¹¹ To illustrate our second result, consider two responders $i \in \{2, 3\}$ having i.i.d. acceptance costs

$$c_i = \begin{cases} 0.20 & \text{with prob. } 0.65, \\ 0.45 & \text{with prob. } 0.35, \end{cases}$$

Table 2 reports the success probabilities, corresponding transfers, and proposer payoffs under unanimity and majority for all candidate proposals. We highlight the equilibrium

¹¹ Of course, there are also cases where both rules lead to the same or opposite qualitative outcomes, as we will see later.

payoffs, transfers, and probabilities of success in bold. Under unanimity, reaching certainty would require $(\bar{c}, \bar{c}) = (0.45, 0.45)$, which is very expensive (total transfers 0.90 and payoff only 0.10). Because the low cost $\underline{c} = 0.20$ is already quite high, the proposer prefers to save on transfers and offer $(0.20, 0.20)$, accepting a substantial risk of failure (success probability of 0.42) for a payoff of 0.254. By contrast, under majority it is optimal to play safe with the high transfer: moving from $(0.20, 0.20)$ to $(0, 0.45)$ slightly increases transfers (from 0.40 to 0.45) but delivers certainty instead of 88% success, raising the proposer's expected payoff from 0.526 to 0.550. Thus, when "low" types are not very cheap, majority rule leads the proposer to secure one decisive vote with a large transfer, while under unanimity she settles for two relatively costly low transfers and a lower success probability – implying higher transfers under majority for $(\underline{c}, \bar{c}, p) = (0.20, 0.45, 0.35)$.

4 Model

We consider a committee of three players who must decide on the allocation of a divisible resource, normalized to size one using a pre-specified voting rule denoted by $r \in \{u, m\}$, where u stands for unanimity and m for majority. The game follows an ultimatum bargaining structure: a *proposer*, indexed as player $i = 1$, submits a proposal specifying a transfer $t_i \geq 0$ to each of the two *responders*, indexed by $i \in \{2, 3\}$. Each responder i has a privately known acceptance cost $c_i \in \{\underline{c}, \bar{c}\}$. The acceptance cost has to be borne by player i , if the proposed measure is adopted.¹² The acceptance cost defines the minimum transfer required for accepting the proposal where $0 \leq \underline{c} < \bar{c}$.

After observing the proposal, the responders vote simultaneously to accept or reject. We denote responder i 's voting decision by $a_i \in \{0, 1\}$, where $a_i = 1$ indicates acceptance and $a_i = 0$ rejection. The proposal is implemented if it meets the required voting threshold. Under *majority rule*, the proposal is accepted if at least one of the two responders votes in favor. Under *unanimity rule*, both responders must vote in favor for the proposal to be accepted. If the proposal is rejected, the disagreement payoff is zero. If the proposal is accepted, the proposer retains the remaining portion of the resource, so that the payoffs are

$$U_1 = 1 - (t_2 + t_3), \quad U_2 = t_2 - c_2, \quad U_3 = t_3 - c_3.$$

To ensure feasibility, the transfers must satisfy the constraint

$$t_2 + t_3 \leq 1.$$

It is straightforward that if the acceptance costs of the responders are commonly observed and the reform is efficient (i.e., $c_2 + c_3 \leq 1$), the reform would be approved in equilibrium, and the total transfers would be weakly higher under the unanimity rule than under the majority rule. Indeed, if $c_2 \leq c_3$, the proposer would offer total transfers equal to $c_2 + c_3$ under unanimity, but only equal to c_2 under majority. Moreover, if the reform is inefficient

¹²The acceptance cost only occurs when the measure is implemented. The vote itself does not create a political or economic cost to the responder. If the responder casts an acceptance vote but the proposal is rejected in a unanimity system, the responder bears no cost.

(i.e., $c_2 + c_3 > 1$), the reform is never approved under unanimity, but it is approved under majority if $c_2 \leq 1$. This leads to the following benchmark observation.

Observation *When acceptance costs are publicly observable, then, in equilibrium, the probability of reform approval is weakly higher under majority, and the total transfers associated with a reform that is approved with positive probability are weakly higher under unanimity.*

In our model, there is incomplete information regarding the responders' acceptance costs. The acceptance cost is drawn independently across responders according to a Bernoulli distribution, $P(c_i = \bar{c}) = p$ with $p \in (0, 1)$.¹³ The proposer does not observe the realization of (c_2, c_3) but holds beliefs given by the common prior and aims to maximize her expected payoff given the feasibility constraint. Given that an accepted proposal yields $1 - (t_2 + t_3)$ and a rejected proposal yields zero, the expected payoff under voting rule $r \in \{u, m\}$ is

$$\mathbb{E}[U_1] = [1 - (t_2 + t_3)] P_r(A).$$

where $P_r(A)$ is the probability of agreement under voting rule r . We analyze equilibrium behavior using the concept of Perfect Bayesian Equilibrium.

Throughout, we focus on efficient reforms ($0 \leq \underline{c} < \bar{c} \leq \frac{1}{2}$), i.e. cases in which implementation maximizes the sum of payoffs.¹⁴ Total payoffs from implementation equal

$$W = [1 - (t_2 + t_3)] + (t_2 - c_2) + (t_3 - c_3) = 1 - (c_2 + c_3).$$

Implementation is (ex post) efficient iff $c_2 + c_3 \leq 1$. Under unanimity, any profile with $c_2 + c_3 > 1$ is never adopted in equilibrium because feasibility ($t_2 + t_3 \leq 1$) and individual rationality ($t_i \geq c_i$ to induce acceptance) jointly fail. Hence, unanimity screens out inefficient reforms by construction. By focusing on efficient reforms, we isolate the informational and strategic differences between the two rules and disregard the question of whether either rule selects an efficient or inefficient outcome.

5 Equilibrium analysis

In the next subsections, we characterize the set of situations (i.e., configurations of preferences and beliefs), in which each type of proposal is selected in equilibrium, and we analyze how these results vary with respect to changes in the key parameters and the rule under consideration. Formally we define the equilibrium as follows:

¹³ Our analysis and main results are robust when responders face different probabilities of incurring the high cost; we impose equal probabilities only for simplicity. The results are available from the authors upon request.

¹⁴ Since our aim is to investigate whether there are cases in which proposers might benefit from unanimity, and responders from majority rule, it is reasonable to limit attention to reforms that are welfare improving. One can easily extend the analysis to potentially welfare-reducing reforms without providing additional substantial insights.

Definition 1 An equilibrium of this game under voting rule $r \in \{u, m\}$ is a *Perfect Bayesian Equilibrium (PBE)* consisting of:

1. A proposal (t_2^*, t_3^*) by the proposer that maximizes her expected payoff,
2. The voting strategies for each responder $i \in \{2, 3\}$ such that

$$a_i(c_i, t_i^*) = \begin{cases} 1, & \text{if } t_i^* \geq c_i, \\ 0, & \text{otherwise,} \end{cases}$$

3. Beliefs consistent with Bayes' rule whenever possible.

Given these strategies and beliefs, the proposer correctly anticipates the probability of approval $P_r(A)$ implied by the responders' acceptance decisions and chooses (t_2^*, t_3^*) satisfying the feasibility constraint $t_2 + t_3 \leq 1$.

5.1 Unanimity

We start the analysis considering the unanimity rule. Note that the offers in the interval $(\underline{c}, \bar{c})$ reduce the proposer's payoff without increasing the probability of acceptance. This is why we restrict the strategy space to transfers in $\{0, \underline{c}, \bar{c}\}$ since any other transfer is strictly dominated.

Proposition 1 *Each of the following three transfer profiles is strictly optimal in a nonempty region of the parameter space $(\underline{c}, \bar{c}, p) \in [0, \frac{1}{2}) \times (\underline{c}, \frac{1}{2}] \times (0, 1)$. In particular:*

1. For $p < \frac{\bar{c}-\underline{c}}{1-2\underline{c}}$, the transfer proposal $t = (\underline{c}, \underline{c})$ is strictly optimal with probability of agreement $P_u(A) = (1-p)^2$.
2. For $\frac{\bar{c}-\underline{c}}{1-2\underline{c}} < p < \frac{\bar{c}-\underline{c}}{1-\underline{c}-\bar{c}}$, the transfer proposal $t = (\bar{c}, \underline{c})$ is strictly optimal with probability of agreement $P_u(A) = 1-p$.
3. For $p > \frac{\bar{c}-\underline{c}}{1-\underline{c}-\bar{c}}$, the transfer $t = (\bar{c}, \bar{c})$ is strictly optimal with probability of agreement $P_u(A) = 1$.

Proof All proofs are in Appendix A. □

Figure 1 illustrates how the threshold curves p_{12}, p_{13}, p_{23} partition the $(p, \underline{c})$ -plane.

In Figure 2, we present the regions where each proposal is optimal across the ranges of p (horizontal axis) and $\underline{c}$ (vertical axis) for two different levels of $\bar{c}$. The green region corresponds to the high transfer proposal, $t = \{\bar{c}, \bar{c}\}$, the yellow region shows where the mixed proposal $t = \{\underline{c}, \bar{c}\}$ is optimal, and the blue region indicates the low transfer proposal $t = \{\underline{c}, \underline{c}\}$. When the likelihood of encountering high-cost responders is low (low p), as in routine regulatory or technical harmonization, the proposer can afford to make minimal offers, expecting broad alignment. This “aggressive” strategy maximizes the proposer's surplus but risks occasional failure if opposition is underestimated. When the chances for a responder increase (medium p), the proposer makes a generous offer to one responder and a low offer to the other. Who receives the high offer is arbitrary, as the proposer has no

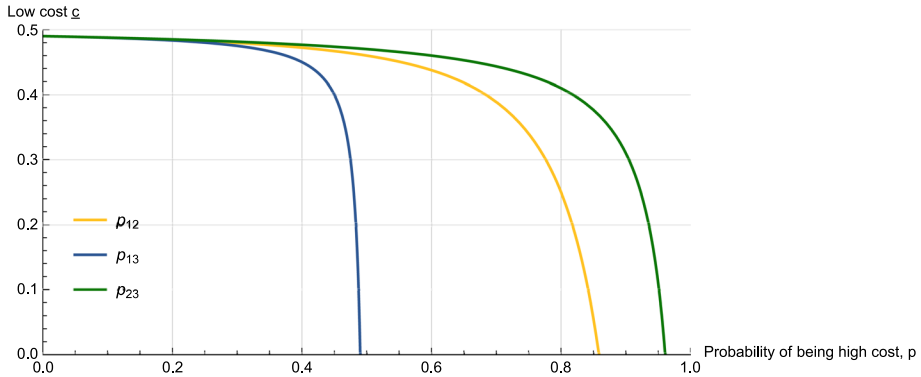


Fig. 1 Probability thresholds separating optimal unanimous offers. The curves plot the cutoff values p_{12} , p_{13} and p_{23} implied by Eqs. (6), (7) and (8) in Appendix A as functions of the low cost $\bar{c}$, for a fixed high cost $\bar{c} = 0.49$. Below p_{12} , the proposer prefers $(\underline{c}, \underline{c})$ over $(\bar{c}, \bar{c})$. Below p_{13} , $(\underline{c}, \underline{c})$ is better than $(\bar{c}, \underline{c})$; and below p_{23} , $(\bar{c}, \underline{c})$ dominates $(\bar{c}, \bar{c})$ for the proposer

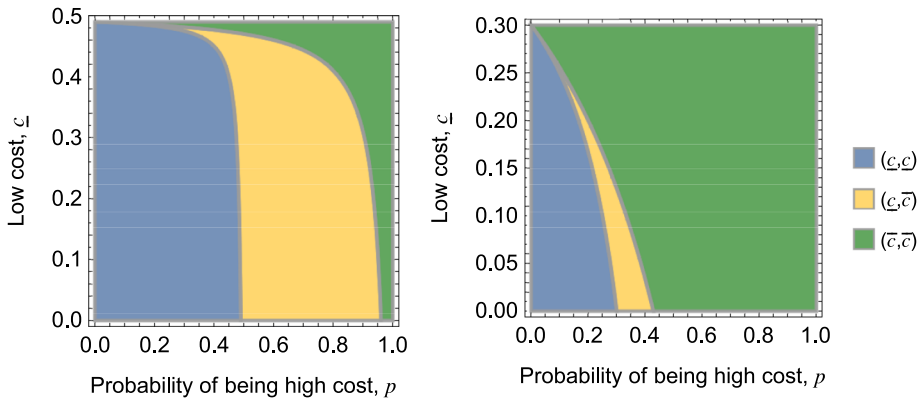


Fig. 2 Optimal unanimous offers as a function of the probability p that a responder is high-cost (horizontal axis) and the low cost $\bar{c}$ (vertical axis). Each colored region indicates which unanimous offer is optimal. The left (right) panel fixes the high cost at $\bar{c} = 0.49$ ($\bar{c} = 0.30$)

differential information on the two responders. However, "a bit of gambling" is better than the aggressive strategy, which would fail too often, and the safe strategy, which would be too expensive. When most member states are expected to have high reservation costs (high p) – as in many high-stakes negotiations – the proposer adopts a “safe” strategy, offering substantial concessions to all. This ensures agreement but at the cost of high total transfers and reduced proposer rents. The economic driver is the “certainty premium”: the cost of guaranteeing reform passage in the face of widespread, but opaque, domestic constraints.

5.1.1 Comparative statics

How do changes in $\bar{c}$ affect the optimal transfers? Note from Fig. 2, that as $\bar{c}$ gets small – and closer to $\underline{c}$ – the mixed transfer proposal $\{\underline{c}, \bar{c}\}$ region shrinks, while the region where $(\bar{c}, \bar{c})$

is optimal expands. Formally, the movements of thresholds are described by the first partial derivatives of the thresholds with respect to $\bar{c}$.

$$\begin{aligned}\frac{\partial p_{12}}{\partial \underline{c}} &= \frac{2\bar{c} - 1}{(2\underline{c} - 1)^2 \sqrt{\frac{2\bar{c}-1}{2\underline{c}-1}}} < 0, & \frac{\partial p_{12}}{\partial \bar{c}} &= -\frac{1}{(2\underline{c} - 1) \sqrt{\frac{2\bar{c}-1}{2\underline{c}-1}}} > 0 \\ \frac{\partial p_{13}}{\partial \underline{c}} &= \frac{1}{2\underline{c} - 1} - \frac{2(\underline{c} - \bar{c})}{(2\underline{c} - 1)^2} < 0, & \frac{\partial p_{13}}{\partial \bar{c}} &= -\frac{1}{2\underline{c} - 1} > 0 \\ \frac{\partial p_{23}}{\partial \underline{c}} &= -\frac{\underline{c} - \bar{c}}{(\bar{c} + \underline{c} - 1)^2} + \frac{1}{\bar{c} + \underline{c} - 1} < 0, & \frac{\partial p_{23}}{\partial \bar{c}} &= -\frac{\underline{c} - \bar{c}}{(\bar{c} + \underline{c} - 1)^2} - \frac{1}{\bar{c} + \underline{c} - 1} > 0\end{aligned}$$

In Fig. 2, we can see each curve dropping steeply as $\bar{c}$ approaches the upper boundary $\underline{c}$ and the yellow regions that represent “mixed” proposals vanish near the top edge (where $\bar{c}$ is close to $\underline{c}$). When $\bar{c}$ falls, even the “high-cost” responder becomes cheap to buy off, so the proposer has less need to hedge against the possibility of facing a genuinely high-cost type. Geometrically, this pushes the threshold curves to the left in the $(p, \underline{c})$ -plane because the proposer can afford to make higher offers over a wider range of probabilities p . The difference between paying a “low”-type cost versus a “high”-type cost is negligible, so the proposer is almost indifferent about whom to compensate. Hence, one needs a smaller value of p (that is, a lower perceived likelihood of encountering a truly high-cost type) before it becomes worthwhile to make a safer (but more expensive) offer. In the figure, this is presented as a leftward shift of the boundary curves, expanding the region of “high-offer” strategies to higher probabilities.

5.2 Majority rule

We proceed with the majority rule and obtain the equilibrium configuration under this voting rule. The optimal offers are as described by Proposition 2.

Proposition 2 *There exists a unique threshold $c^*(\bar{c}) > 0$, such that each of the following three transfer profiles is strictly optimal in a nonempty region of the parameter space $(\underline{c}, \bar{c}, p) \in [0, \frac{1}{2}] \times (0, \frac{1}{2}] \times (0, 1)$. In particular:*

1. *If $\underline{c} < c^*$, there are three distinct intervals in p such that:*
 - i. *For $0 < p \leq \frac{\underline{c}}{1 - 2\underline{c}}$, the optimal transfer is $t = (0, \underline{c})$, with agreement probability $1 - p$.*
 - ii. *For $\frac{\underline{c}}{1 - 2\underline{c}} < p \leq \sqrt{\frac{\bar{c} - 2\underline{c}}{1 - 2\underline{c}}}$, the optimal transfer is $t = (\underline{c}, \underline{c})$, with agreement probability $1 - p^2$.*
 - iii. *For $\sqrt{\frac{\bar{c} - 2\underline{c}}{1 - 2\underline{c}}} < p \leq 1$, the optimal transfer is $t = (0, \bar{c})$, with agreement probability 1.*
2. *If $\underline{c} \geq c^*$, there are two distinct intervals in p such that:*

- i. For $0 < p \leq \frac{\bar{c} - \underline{c}}{1 - \underline{c}}$, the optimal transfer is $(0, \underline{c})$, with agreement probability $1 - p$.
- ii. For $\frac{\bar{c} - \underline{c}}{1 - \underline{c}} < p \leq 1$, the optimal transfer is $(0, \bar{c})$, with agreement probability 1.

Proof All proofs are in Appendix A. □

Figure 3 illustrates how the threshold curves $\hat{p}_{12}$, $\hat{p}_{13}$, $\hat{p}_{23}$ partition the $(p, \underline{c})$ -plane.

Figure 4 shows the profit-maximizing configuration of offers for two high cost parameter values. The comparison of the two panels illustrates the changes in the optimal offer regions as the high cost parameter value goes from 0.49 (left) to 0.3 (right). The horizontal axis depicts the probability p of a high-type responder, while the vertical axis displays the low acceptance cost $\underline{c}$, which can range from 0 to $\bar{c}$. Depending on the parameters, each of the three offers can be optimal. In the yellow area, for low values of $\underline{c}$ and intermediate values of p , it is optimal to offer $t = (\underline{c}, \underline{c})$ hoping that one of the responders accepts the low offer. In the green area, when p is high and $\underline{c}$ is also large, the “safe” proposal $t = (0, \bar{c})$ is chosen. Finally, in the blue area, characterized by low probabilities of a high type and intermediate values of $\underline{c}$, the best offer is $t = (0, \underline{c})$. Here, the proposer is gambling on the assumption that the responder, who receives the $\underline{c}$ -offer, is of the low type, as the other responder will reject in any case.

With majority rule, the basic risk-reward trade-offs are similar to those in the unanimity case, but here, the proposer must gain acceptance from only one responder country. This opens up an avenue for the gambling strategy. When the likelihood of high-cost opposition is low, the proposer can gamble by making a low offer to one responder while offering nothing to the other country, betting that the one responder with the low offer will help to secure the majority – a tactic often viable in less contentious EU policy issues where the damage in case of failure is not detrimental. As the probability of encountering high-cost responders increases, the proposer shifts to a more balanced approach, offering moderate transfers to both responders in the hope that at least one will accept, mirroring situations where proposers in the EU make “insurance” offers to potential veto players. Finally, when high-cost

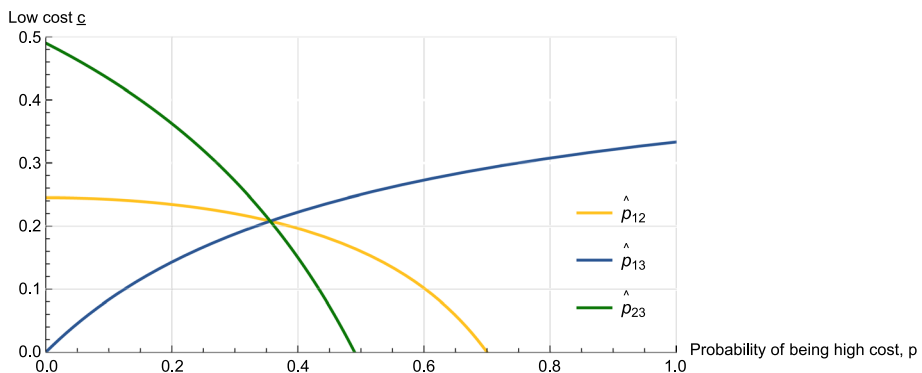


Fig. 3 Probability thresholds separating optimal majority offers. The three curves plot the cutoff values $\hat{p}_{12}$, $\hat{p}_{13}$ and $\hat{p}_{23}$ implied by equations (12), (13) and (14) in the Appendix A as functions of the low cost $\underline{c}$, for a fixed high cost $\bar{c} = 0.49$. Below $\hat{p}_{12}$, the proposer prefers $(\underline{c}, \underline{c})$ over $(0, \bar{c})$. Below $\hat{p}_{13}$, $(\underline{c}, \underline{c})$ is better than $(0, \underline{c})$; and below $\hat{p}_{23}$, $(0, \underline{c})$ dominates $(0, \bar{c})$ for the proposer

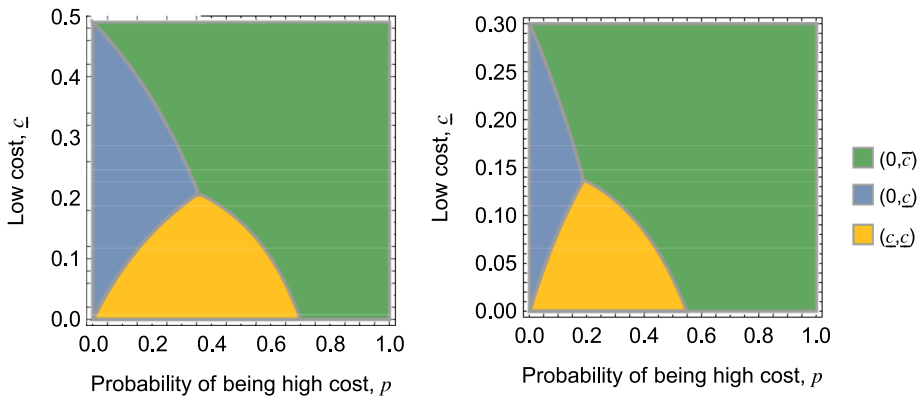


Fig. 4 Optimal majority-rule offers as a function of the probability p that a responder is high-cost (horizontal axis) and the low cost $\underline{c}$ (vertical axis). Each colored region indicates which majority offer is optimal. The left (right) panel fixes the high cost at $\bar{c} = 0.49$ ($\bar{c} = 0.30$)

opposition is substantial, the proposer adopts a safe strategy by offering a large transfer to one responder and zero to the other, ensuring passage but at a higher cost. This reflects how, in high-stakes or politically salient reforms, the proposer may need to focus resources on securing a pivotal member's consent, accepting higher total transfers to guarantee reform approval under majority rule.

Note that for low values of $\underline{c}$, all three proposals can be optimal (depending on p), while only $(0, \underline{c})$ and $(0, \bar{c})$ are candidate proposals for high acceptance costs of the low type. In the following, we will distinguish whether two or three proposals can be optimal. Therefore, we introduce $\underline{c}^*$ as the critical level of acceptance costs distinguishing these two cases.

5.2.1 Comparative statics

The comparative statics are formally captured by the partial derivatives with respect to both $\underline{c}$ and $\bar{c}$. Specifically, we have:

$$\frac{\partial \hat{p}_{12}}{\partial \underline{c}} = \frac{-\frac{2}{1-2\underline{c}} + \frac{2(\bar{c}-2\underline{c})}{(1-2\underline{c})^2}}{2\sqrt{\frac{\bar{c}-2\underline{c}}{1-2\underline{c}}}} < 0, \quad \frac{\partial \hat{p}_{12}}{\partial \bar{c}} = \frac{1}{2(1-2\underline{c})\sqrt{\frac{\bar{c}-2\underline{c}}{1-2\underline{c}}}} > 0, \quad (1)$$

$$\frac{\partial \hat{p}_{13}}{\partial \underline{c}} = \frac{1}{1-2\underline{c}} + \frac{2\underline{c}}{(1-2\underline{c})^2} > 0, \quad \frac{\partial \hat{p}_{13}}{\partial \bar{c}} = 0, \quad (2)$$

$$\frac{\partial \hat{p}_{23}}{\partial \underline{c}} = -\frac{1}{1-\underline{c}} + \frac{\bar{c}-\underline{c}}{(1-\underline{c})^2} < 0, \quad \frac{\partial \hat{p}_{23}}{\partial \bar{c}} = \frac{1}{1-\underline{c}} > 0. \quad (3)$$

The sign of $\frac{\partial \hat{p}_{12}}{\partial \underline{c}}$ and $\frac{\partial \hat{p}_{23}}{\partial \underline{c}}$ indicates that these thresholds are increasing in $\bar{c}$ and hence the curves shift left when $\bar{c}$ decreases, while $\frac{\partial \hat{p}_{13}}{\partial \bar{c}} = 0$ implies that $\hat{p}_{13}$ remains fixed in the horizontal direction. These shifts cause the right-hand region of high transfers, bounded by $\hat{p}_{12}$ and $\hat{p}_{23}$, to expand. This conforms to the economic intuition that the safe offer is made more often if the cost of these offers decreases (lower $\bar{c}$).

6 Comparison

In this section, we compare the acceptance probabilities under majority and unanimity rules across two different scenarios, distinguished by whether the low acceptance cost $\underline{c}$ is greater than or equal to the critical threshold c^* or strictly below it. When $\underline{c} \geq c^*$, we show that unanimity typically requires higher transfers but does not exceed the agreement probability attainable under majority rule. In contrast, when $\underline{c} < c^*$, a “paradoxical” outcome may arise in which unanimity achieves a strictly higher acceptance probability despite imposing a more stringent voting requirement. The following subsections detail these scenarios and illustrate how the proposer’s optimal strategy and overall agreement probability vary in each case.

6.1 High acceptance costs: $\underline{c} \geq c^*$

In this subsection, we examine the case where the low acceptance cost $\underline{c}$ is at least as large as the critical threshold $\underline{c}^*$. Figure 5 shows the optimal offers and corresponding agreement probabilities under unanimity and majority rule. As $\frac{\bar{c}-\underline{c}}{1-\underline{c}} < \frac{\bar{c}-\underline{c}}{1-2\underline{c}}$, there is no parameter value for p where the acceptance probability under unanimity can exceed the acceptance probability under majority rule.

Transfers are generally higher under unanimity compared to majority voting. However, for the parameter range

$$\frac{\bar{c}-\underline{c}}{1-\underline{c}} < p < \frac{\bar{c}-\underline{c}}{1-2\underline{c}}, \quad (4)$$

the proposer actually pursues an aggressive strategy under unanimity by offering $(\underline{c}, \underline{c})$, which totals $2\underline{c}$. Under the majority rule, in contrast, she adopts a safe strategy by offering $(0, \bar{c})$, which has a total transfer of $\bar{c}$. Thus, whenever $\bar{c} > 2\underline{c}$, the total cost under majority is higher than under unanimity in that range.

Numerical example. As in the example in Section 3, let $\underline{c} = 0.20$ and $\bar{c} = 0.45$. The bounds in (4) become

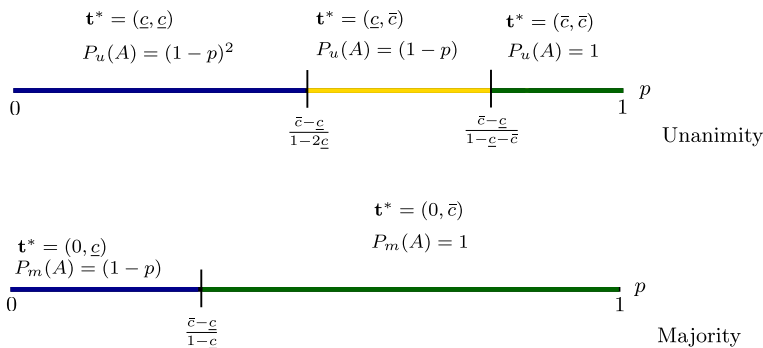


Fig. 5 Comparison of Agreement Probabilities and Optimal Offer of Transfers under High Acceptance Costs, $\underline{c} \geq c^*$

$$\frac{\bar{c} - \underline{c}}{1 - \underline{c}} = 0.3125, \quad \frac{\bar{c} - \underline{c}}{1 - 2\underline{c}} = 0.4167.$$

Hence, for

$$0.3125 < p < 0.4167,$$

the proposer offers $(\underline{c}, \underline{c}) = (0.20, 0.20)$ under unanimity and $(0, \bar{c}) = (0, 0.45)$ under majority rule. Since $0.45 > 2 \cdot 0.20$, total transfers are higher under majority in this range.

This observation is summarized in Remark 1:

Remark 1 For a certain range of probability values p (see (4)), the total transfer under majority rule exceeds that under unanimity for $\bar{c} > 2\underline{c}$.

Despite this difference in transfers, the unanimity rule never achieves a higher acceptance probability: paying both responders under unanimity does not yield an agreement rate that beats the majority rule, which can rely on securing just one responder's vote.

6.2 Low acceptance costs: $\underline{c} < c^*$

We now turn to the case of low acceptance costs of the low type. In this setting, the unanimity rule always involves higher transfers than the majority rule. Hence, our comparison will focus on differences in agreement probabilities rather than expenditures. Figure 6 illustrates the optimal offers and resulting probabilities of acceptance for both rules when the low type's acceptance cost $\underline{c}$ is small. Here, the ordering of equilibrium configurations is less clear. In many parameter configurations, the acceptance probabilities are either identical under the two rules or higher under the majority rule – particularly when p is small, since $(1 - p)^2 < 1 - p$.

The interesting cases are those where the probability of successful offers is higher under unanimity than under majority. Figure 7 illustrates such a case. Formally, this occurs if

$$\frac{\bar{c} - \underline{c}}{1 - \underline{c} - \bar{c}} < \sqrt{\frac{\bar{c} - 2\underline{c}}{1 - 2\underline{c}}},$$

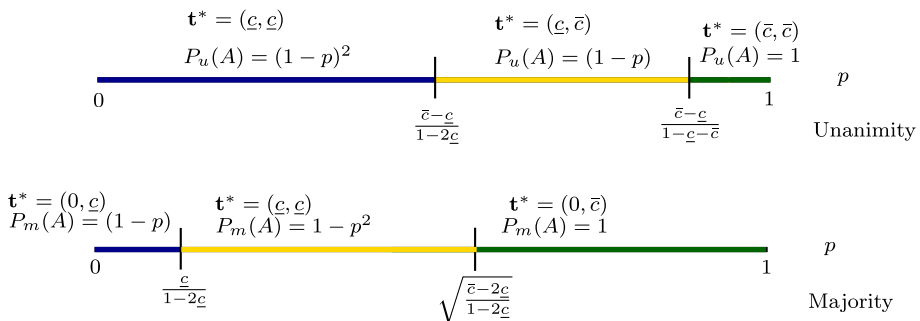


Fig. 6 Comparison of Agreement Probabilities with Low Acceptance Costs, $\underline{c} < c^*$

so that, whenever this holds, for some nonempty interval of p , the majority-rule equilibrium achieves only $1 - p^2$ agreement probability while acceptance is certain under unanimity rule.

A straightforward way to illustrate this “paradoxical” outcome is to consider the special case $\underline{c} = 0$. In that instance, the relevant inequality reduces to $1 - \bar{c} > \sqrt{\bar{c}}$, which is satisfied for $\bar{c} < 0.382$. Thus, whenever $0 \leq \bar{c} < 0.382$, there are probability values $\frac{\bar{c}}{1-\bar{c}} < p < \sqrt{\bar{c}}$ for which unanimity achieves a higher acceptance rate ($= 1$) than majority rule ($= 1 - p^2$). The parameter range for which the ‘paradoxical’ result can hold is given by $\frac{\bar{c}-\underline{c}}{1-\underline{c}-\bar{c}} < \sqrt{\frac{\bar{c}-2\underline{c}}{1-2\underline{c}}}$ for $\underline{c}$ and we obtain

$$\underline{c} < \frac{1}{7\bar{c}-3} [-(2\bar{c}-1)^2 - (2\bar{c}-1)\sqrt{(2\bar{c}-1)(\bar{c}-1)} - 2\bar{c} + 5\bar{c}^2] \quad (5)$$

This condition is depicted in Fig. 8. The acceptance costs of the high type are plotted on the horizontal axis, the acceptance costs of the low type on the vertical axis. The blue area shows those parameter values where there are probabilities p so that the unanimity rule leads to an acceptance of an optimal proposal with certainty, while the acceptance remains uncertain with the majority rule.¹⁵

Numerical example. Take $\underline{c} = 0.05$ and $\bar{c} = 0.2$ as in Section 3. Under unanimity, the proposer prefers the safe high offer $(\bar{c}, \bar{c}) = (0.2, 0.2)$ if the expected payoff $1 - 2\bar{c}$ weakly exceeds that of the risky alternatives; this occurs for

$$p > \frac{\bar{c} - \underline{c}}{1 - \underline{c} - \bar{c}} = \frac{0.15}{0.75} = 0.2.$$

Under majority, the risky low offer $(\underline{c}, \underline{c})$ is optimal when it beats both the safe $(0, \bar{c})$ and the asymmetric $(0, \underline{c})$ options, which yields the interval

$$\frac{\underline{c}}{1-2\underline{c}} < p < \sqrt{\frac{\bar{c}-2\underline{c}}{1-2\underline{c}}} \iff \frac{0.05}{0.9} < p < \sqrt{\frac{0.1}{0.9}} \approx 0.0556 < p < 0.3333.$$

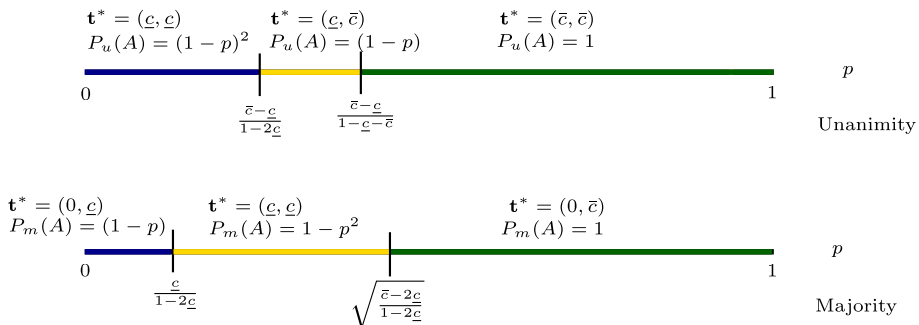
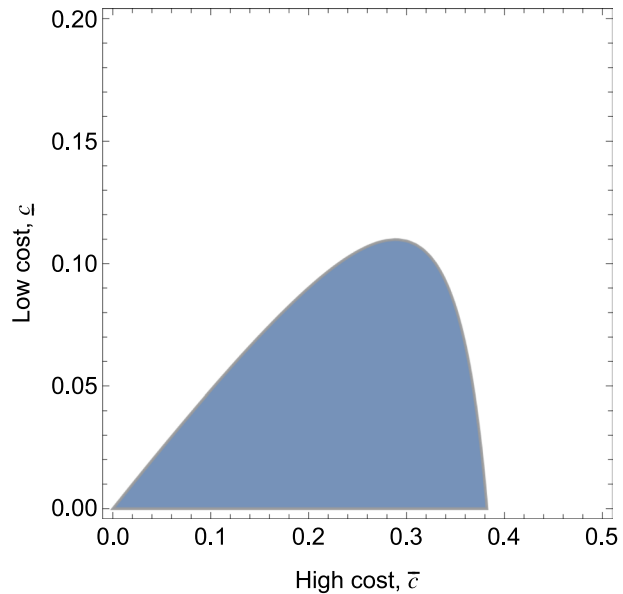


Fig. 7 Comparison of agreement probabilities with low acceptance costs

¹⁵ It can easily be checked that the other conditions $-\underline{c} < \bar{c}$ and $\underline{c} < \underline{c}^*$ are fulfilled in the relevant range.

Fig. 8 Parameter values of low and high acceptance costs under which unanimity leads to higher acceptance rates than majority rule



Hence, for any $p \in (0.2, 0.3333)$ the two rules prescribe different best responses: unanimity selects $(\bar{c}, \bar{c})$, while majority keeps gambling with $(\underline{c}, \underline{c})$.

As we showed in Sect. 3, with $(\underline{c}, \bar{c}, p) = (0.05, 0.2, 0.25)$, unanimity implements with certainty while majority succeeds with probability $1 - p^2 < 1$, delivering the paradox: acceptance is (strictly) higher under unanimity. Under majority, moving from the risky profile $(\underline{c}, \underline{c})$ to the safe profile $(0, \bar{c})$ buys success only in the “both would otherwise reject” states: the success gain is $p^2 = 0.0625$, while the transfer cost rises by $\bar{c} - 2\underline{c} = 0.1$. Under unanimity, insurance is stronger: raising one offer from $\underline{c}$ to $\bar{c}$ (going from $(\underline{c}, \underline{c})$ to $(\underline{c}, \bar{c})$) increases success probability by $p(1 - p) = 0.25 \times 0.75 = 0.1875$ at cost $\bar{c} - \underline{c} = 0.15$; raising the second offer (from $(\underline{c}, \bar{c})$ to $(\bar{c}, \bar{c})$) adds a further success gain of $p = 0.25$ for another $\bar{c} - \underline{c} = 0.15$. In short, the safe move under majority rule removes relatively few failure states, whereas the insurance steps under unanimity remove many more. Hence, within the parameter range above, unanimity increases the likelihood of approval, even though its approval threshold is stricter.

The key insights from the above discussion are summarized in Remark 2:

Remark 2 There exists a critical threshold c^{**} satisfying $c^{**} < c^*$, such that for every $\underline{c} < c^{**}$, there exists an interval of probability values p where the unanimity rule guarantees the acceptance of an optimal proposal with certainty, whereas under the majority rule, acceptance remains uncertain.

The underlying intuition is that when the proposer considers switching from a risky offer to a safe one, the trade-off differs between the two voting rules. Under majority rule, the switching increases the costs by $\bar{c} - 2\underline{c}$, but the probability of acceptance rises by p^2 . Under unanimity, the cost increases by $\bar{c} - \underline{c}$, while the probability increases by p . This means that while the cost of moving to the safe option is lower under majority rule, an effect that

becomes more pronounced as $\underline{c}$ increases, the corresponding benefit in terms of a higher probability of acceptance is greater under unanimity.

These findings are not an artifact of our baseline assumptions. In the next section, we discuss that the key mechanisms remain robust to richer cost distributions, larger committees, and the introduction of pre-offer cheap talk communication.

7 Extensions and discussion

7.1 Cost assumptions

We assume binary acceptance costs – either a “low” $\underline{c}$ or a “high” $\bar{c}$ – because this benchmark cleanly captures the insurance-versus-gambling trade-off and keeps the proposer’s candidate offers transparent. This is our baseline for tractability. However, the phenomenon is not an artifact of two support points: with three cost types, the same logic survives and generates a region in which unanimity delivers a higher passage probability than majority (see Appendix B, Table 3, 4 and 5 for a numerical confirmation). The intuition also extends to continuous environments: a bimodal cost distribution – one low-cost mode and a high-cost mode – preserves the same incentives, pushing unanimity toward broad “insurance” and the majority toward “gambling,” thereby sustaining the paradox beyond the binary benchmark. By contrast, in a unimodal (single-peaked) distribution concentrated around an intermediate cost level, there is no distinct mass of very high-cost types to insure against. In such cases, both rules lead the proposer to target essentially the same intermediate group of responders, so the scope for divergence in success probabilities becomes very limited.¹⁶

7.2 Multiple responders

The mechanism behind the paradox is not tied to having only two responders. In the binary benchmark, with n responders whose costs are independently drawn from a common distribution over $\{\underline{c}, \bar{c}\}$, the proposer faces the same basic trade-off between a risky low offer and a safe high offer. Under unanimity, a low offer succeeds only if all n responders have low costs; thus, the set of failure states is large, as any realization with at least one high-cost responder leads to rejection. Moving from the low to the high offer eliminates the risk of failure.

Under majority, by contrast, the low offer already passes in all states with “enough” low-cost types (at least a majority of responders), so the set of failure states is much smaller. Switching to the safe high offer only buys success in those relatively rare states where too many responders happen to be high-cost, while the extra transfers grow linearly with committee size. For suitable parameters, the proposer therefore insures under unanimity but continues to gamble under majority.

As a simple illustration beyond the two-responder case, consider three responders with i.i.d. costs $c_i \in \{0.05, 0.25\}$ and $\Pr(c_i = 0.05) = 0.55$. Table 6 in Appendix B reports the implementation probabilities and expected payoffs for several candidate

¹⁶ For a bargaining approach with incomplete information and continuous reservation prices, see, e.g., Konrad and Thum (2021).

offers. Under unanimity, the optimal offer is $(0.25, 0.25, 0.25)$, which guarantees success with probability 1 and yields a payoff of 0.25. Under majority rule, instead, the proposer optimally chooses the asymmetric offer $(0.05, 0.05, 0.25)$ (up to permutation), which yields an expected payoff of 0.518 under majority, but success is uncertain (success probability = 0.798). Hence, in this three-responder example, unanimity again delivers a strictly higher acceptance probability than majority rule at the respective optima.

7.3 Pre-offer cheap talk communication

The key ingredient for our paradoxical result is the proposer's ex ante uncertainty about the responders' acceptance costs. We therefore work in a static environment with imperfect information and abstract from repeated interaction or dynamic learning, which would gradually eliminate this uncertainty. A natural question is whether allowing for a pre-offer cheap talk stage between the proposer and the responders could eliminate the paradox. We examine this question only within the parameter region in which the paradox arises in our baseline model.

To test the robustness of the paradox – namely, the range of parameters for which unanimity yields a higher acceptance probability than majority – we extend the baseline model by introducing a pre-offer communication stage. We investigate whether a fully informative cheap-talk equilibrium can be sustained under majority rule within the previously identified paradoxical interval. We use the concrete numerical example from the previous subsection, with low acceptance cost $\underline{c} = 0.1$, high acceptance cost $\bar{c} = 0.3$, and high-type probability $p \in (0.3333, 0.3536)$. For simplicity, we set $p = 0.34$.

We now analyze whether a fully separating Perfect Bayesian Equilibrium (PBE) can be sustained. In such an equilibrium, both responders would truthfully report their types and the proposer would condition her offer on the received messages.

Consider the timing of the extended game:

0. Nature assigns a cost $c_i \in \{\underline{c}, \bar{c}\}$ to each responder $i \in \{2, 3\}$. Each responder privately observes his own cost.
1. Responders 2 and 3 simultaneously send a costless, non-binding message $m_i \in \{\text{"low"}, \text{"high"}\}$.
2. The proposer observes (m_2, m_3) and chooses transfers (t_2, t_3) .
3. Responders vote to accept or reject.
4. Payoffs are realized according to the majority or unanimity rule.

We begin with majority rule and check whether a fully separating PBE exists. Without loss of generality, consider responder 2. He knows his own cost c_2 , but remains uncertain about responder 3's cost, which is $\bar{c}$ with probability $p = 0.34$ and $\underline{c}$ with probability $1 - p = 0.66$.

We first analyze the incentives of a low-cost responder ($c_2 = 0.1$). We compare his expected payoff from sending a truthful message $m_2 = \text{"low"}$ with the payoff from lying and sending $m_2 = \text{"high"}$.

If responder 2 sends $m_2 = \text{"low"}$, the proposer's response depends on responder 3's type. When responder 3 is low-cost (probability 0.66), the proposer receives (low, low) and secures one vote by offering $(t_2, t_3) = (0.1, 0)$ or $(0, 0.1)$ with equal probability. If

responder 2 receives $t_2 = 0.1$, he obtains 0; if he receives 0, he obtains -0.1 . His expected payoff is therefore -0.05 . When responder 3 is high-cost (0.34), the proposer observes (low, high) and buys responder 2's vote via $(t_2, t_3) = (0.1, 0)$, yielding payoff 0. The combined expected payoff from truth-telling is

$$\mathbb{E}[U_2 \mid m_2 = \text{"low"}] = (1 - p)(-0.05) + p \cdot 0 = -\mathbf{0.033}.$$

If instead responder 2 lies and sends $m_2 = \text{"high"}$, then when responder 3 is low-cost (probability 0.66), the proposer offers $(t_2, t_3) = (0, 0.1)$, inducing a payoff of -0.1 . When responder 3 is high-cost (0.34), the proposer randomizes between (0.3, 0) and (0, 0.3), giving responder 2 an expected payoff of 0.05. Hence, lying yields

$$\mathbb{E}[U_2 \mid m_2 = \text{"high"}] = (1 - p)(-0.1) + p(0.05) = -\mathbf{0.049}.$$

Since $-0.033 > -0.049$, low-cost responders strictly prefer truth-telling.

Turning to high-cost responders ($c_2 = 0.3$), assume first they truthfully send $m_2 = \text{"high"}$. When responder 3 is low-cost (0.66), the proposer buys responder 3's vote with (0, 0.1), giving responder 2 payoff -0.3 . When responder 3 is high-cost (0.34), the proposer randomizes between (0.3, 0) and (0, 0.3), yielding expected payoff -0.15 . Hence

$$\mathbb{E}[U_2 \mid m_2 = \text{"high"}] = (1 - p)(-0.3) + p(-0.15) = -\mathbf{0.249}.$$

If a high-cost responder lies and sends $m_2 = \text{"low"}$, then when responder 3 is low-cost, the proposer mixes between (0.1, 0) and (0, 0.1), leaving responder 2 with expected payoff -0.15 . When responder 3 is high-cost, the proposer offers (0.1, 0), and the policy fails, yielding payoff 0. Therefore

$$\mathbb{E}[U_2 \mid m_2 = \text{"low"}] = (1 - p)(-0.15) + p \cdot 0 = -\mathbf{0.099}.$$

Since $-0.099 > -0.249$, high-cost responders strictly prefer to lie.

Thus, no fully separating PBE exists under majority rule within the paradoxical parameter range. Low-cost types are truthful, but high-cost types strictly prefer to misreport. Because high-cost types induce rejection when misclassified, cheap talk cannot sustain informative communication.

Under unanimity, consider the following: irrespective of their types, both responders always send $m_i = \text{"high"}$. The proposer ignores the messages and plays the unanimity strategy from the baseline model. This defines a trivial pooling PBE in which all types report "high" and cheap talk carries no information. Implementation probabilities exactly coincide with those in the no-communication game.

Combining both voting rules, we conclude that introducing pre-offer cheap talk does not overturn the paradox. Within the paradoxical interval for p , no separating equilibrium exists, and the only pure-strategy equilibria are pooling (babbling) equilibria. These equilibria replicate the baseline outcomes exactly. Hence, for the specific region in which unanimity outperforms majority in acceptance probability, the paradox is fully robust to the introduction of cheap talk. We do not attempt to characterize mixed-strat-

egy equilibria, since the existence of a babbling PBE is sufficient for our robustness result.

8 Conclusion

Our analysis challenges conventional wisdom by illustrating how incomplete information alters the relative efficiency of unanimity and majority voting rules. The proposer's strategic offers under each rule drive key differences in outcomes. Under unanimity, the proposer must secure consent from every responder, which often entails offering larger transfers to ensure approval. Under majority rule, by contrast, the proposer only needs to convince a subset of responders, creating incentives to risk offering smaller transfers. Contrary to the notion that unanimity always forces the proposer to pay more, we identify scenarios where offering modest transfers to both responders under unanimity can be cheaper than concentrating a larger transfer on just one responder under majority rule.

Moreover, although unanimity has a stricter approval threshold, it can still yield a higher probability of reform acceptance. The paradox emerges from the proposer's response to uncertainty about responders' reservation values. Under unanimity, a single refusal defeats the proposal, so the proposer chooses an insurance solution by offering sufficiently large transfers to all responders, eliminating many of the states that would otherwise lead to failure, especially the common case where one responder would reject. Under majority rule, by contrast, the proposer often gambles with lower transfers. Moving from two small offers to one big offer resolves the relatively rare case where both would be rejected. Moving to two high offers yields a modest gain in success probability relative to the extra cost. As a result, the same additional transfers under unanimity remove more failure states and deliver a larger reduction in rejection risk, making unanimity more likely to pass the reform than gambling under majority.

These insights contribute to the ongoing debate on EU decision-making reforms. The conventional view that unanimity necessarily empowers responders and majority rule benefits proposers is not universally valid. Instead, the effects of each rule depend on the level of uncertainty, the distribution of acceptance costs, and the way bargaining unfolds under different voting mechanisms.

A Appendix

Proof of Proposition 1 A responder accepts a proposed transfer t_i if and only if $t_i - c_i \geq 0 \Rightarrow t_i \geq c_i$. Since c_i is binary, the probability of acceptance by responder i is

$$\alpha(t_i) = \begin{cases} 1, & t_i \geq \bar{c}, \\ 1 - p, & \underline{c} \leq t_i < \bar{c}, \\ 0, & 0 \leq t_i < \underline{c}. \end{cases}$$

Under unanimity rule, both responders must vote in favor, leading to an implementation probability of

$$P_u(A) = \alpha(t_2)\alpha(t_3).$$

Let U_1, U_2, U_3 denote the proposer's expected payoffs for three candidate transfer profiles under unanimity. Since offers in the interval $(\underline{c}, \bar{c})$ reduce the proposer's payoff without increasing the probability of acceptance, we restrict the strategy space to transfers in $\{0, \underline{c}, \bar{c}\}$ since any other transfer is strictly dominated. Table 3 displays the three relevant transfer vectors t , the agreement probability and the proposer's payoff in case of agreement.

Table 3 Payoffs and Probabilities with Unanimity Voting

Label	Transfers	Acceptance probability	Proposer's payoff in case of acceptance
U_1	$(\underline{c}, \underline{c})$	$(1 - p)^2$	$1 - 2\underline{c}$
U_2	$(\bar{c}, \bar{c})$	1	$1 - 2\bar{c}$
U_3	$(\underline{c}, \bar{c})$	$(1 - p)$	$1 - \underline{c} - \bar{c}$

Pairwise expected payoff comparisons yield the following threshold conditions:

$$U_1 \geq U_2 \iff 1 - \sqrt{\frac{1 - 2\bar{c}}{1 - 2\underline{c}}} \geq p, \quad (6)$$

$$U_1 \geq U_3 \iff \frac{\bar{c} - \underline{c}}{1 - 2\underline{c}} \geq p, \quad (7)$$

$$U_2 \geq U_3 \iff p \geq \frac{\bar{c} - \underline{c}}{1 - \underline{c} - \bar{c}}. \quad (8)$$

Equations 6 to 8 define the threshold values for p at which the proposer switches between different transfer strategies¹⁷:

$$p_{12} = 1 - \sqrt{\frac{1 - 2\bar{c}}{1 - 2\underline{c}}}, \quad p_{13} = \frac{\bar{c} - \underline{c}}{1 - 2\underline{c}}, \quad p_{23} = \frac{\bar{c} - \underline{c}}{1 - \underline{c} - \bar{c}}.$$

These three thresholds have a clear ordering and hence yield four disjoint nonempty subintervals of $p \in (0, 1)$. Threshold (7) must be to the left of threshold (6): $1 - \sqrt{\frac{1 - 2\bar{c}}{1 - 2\underline{c}}} > \frac{\bar{c} - \underline{c}}{1 - 2\underline{c}}$ boils down to $(\bar{c} - \underline{c})^2 > 0$. Threshold (6) must be to the left of threshold (8): $1 - \sqrt{\frac{1 - 2\bar{c}}{1 - 2\underline{c}}} < \frac{\bar{c} - \underline{c}}{1 - \underline{c} - \bar{c}}$ boils down to $(\bar{c} - \underline{c})^2 < 0$. Moreover, when $\underline{c} = 0$

$$p_{12} = 1 - \sqrt{1 - 2\bar{c}}, \quad (9)$$

$$p_{13} = \bar{c}, \quad (10)$$

$$p_{23} = -\frac{\bar{c}}{\bar{c} - 1}. \quad (11)$$

¹⁷ p_{12} is the threshold from comparing U_1 and U_2 , p_{13} is from comparing U_1 and U_3 , and p_{23} is from comparing U_2 and U_3 .

Since $0 < p_{13} < p_{12} < p_{23} < 1$ (see, Fig. 1), each candidate transfer proposal is strictly optimal for a nonempty region of parameter values. $\square$

Proof of Proposition 2 Similarly to the previous case, a responder accepts a proposed transfer t_i if and only if $t_i - c_i \geq 0 \Rightarrow t_i \geq c_i$. Since c_i is binary, the probability of agreement by responder i is

$$\alpha(t_i) = \begin{cases} 1, & t_i \geq \bar{c}, \\ 1-p, & \underline{c} \leq t_i < \bar{c}, \\ 0, & 0 \leq t_i < \underline{c}. \end{cases}$$

Under majority rule, the proposal is implemented if at least one responder votes in favor, so the probability of implementation is

$$P_m(A) = \alpha(t_2) + \alpha(t_3) - \alpha(t_2)\alpha(t_3).$$

The following table presents the three relevant offers that strictly dominate any other offer under majority rule, their corresponding acceptance probabilities, and the proposer's payoff in case of acceptance (Table 4). Note that $\{\underline{c}, \underline{c}\}$ can dominate $\{0, \bar{c}\}$ if and only if $2\underline{c} < \bar{c} \Rightarrow \underline{c} < \frac{1}{4}$.

Table 4 Payoffs and probabilities with majority voting

Label	Transfers	Acceptance probability	Proposer's payoff in case of acceptance
M_1	$(\underline{c}, \underline{c})$	$1 - p^2$	$1 - 2\underline{c}$
M_2	$(0, \bar{c})$	1	$1 - \bar{c}$
M_3	$(0, \underline{c})$	$1 - p$	$1 - \underline{c}$

Pairwise expected payoff comparisons yield the following threshold conditions:

$$M_1 \geq M_2 \iff p \geq \sqrt{\frac{\bar{c} - 2\underline{c}}{1 - 2\underline{c}}}, \quad (12)$$

$$M_1 \geq M_3 \iff p \geq \frac{\underline{c}}{1 - 2\underline{c}}, \quad (13)$$

$$M_2 \geq M_3 \iff p \geq \frac{\bar{c} - \underline{c}}{1 - \underline{c}}. \quad (14)$$

Equations (12) to (14) define the threshold values for p at which the proposer switches between different transfer strategies.

$$\hat{p}_{12} = \sqrt{\frac{\bar{c} - 2\underline{c}}{1 - 2\underline{c}}}, \quad \hat{p}_{13} = \frac{\underline{c}}{1 - 2\underline{c}}, \quad \hat{p}_{23} = \frac{\bar{c} - \underline{c}}{1 - \underline{c}}.$$

The constraints below define the conditions under which one threshold dominates another:

$$\hat{p}_{12} < \hat{p}_{13} \Rightarrow \left[0 < \underline{c} \leq \frac{1}{6}(3 - \sqrt{3}) \quad \wedge \quad 2\underline{c} \leq \bar{c} < \bar{\bar{c}} \right], \quad (15)$$

$$\hat{p}_{12} > \hat{p}_{23} \Rightarrow \left[0 \leq \underline{c} < \frac{1}{6}(3 - \sqrt{3}) \quad \wedge \quad \bar{\bar{c}} < \bar{c} < 0.5, \right] \quad (16)$$

$$\hat{p}_{13} > \hat{p}_{23} \Rightarrow \left[0 < \underline{c} \leq \frac{1}{6}(3 - \sqrt{3}) \quad \wedge \quad \underline{c} < \bar{c} < \bar{\bar{c}} \right]. \quad (17)$$

The point where the three threshold intersect is given by $c^* = \frac{\bar{c}^2 - 1 - \sqrt{\bar{c}^4 - 3\bar{c}^3 + 4\bar{c}^2 - 3\bar{c} + 1}}{3(\bar{c} - 1)}$ with maximum value $c^* = \frac{1}{6}(3 - \sqrt{3}) \approx 0.21$ when $\bar{c} \rightarrow 0.5$. Moreover, $\bar{\bar{c}} = \frac{3c^{*2} - 2c^*}{2c^* - 1}$ with $\lim_{\underline{c} \rightarrow 0.21} \frac{3c^{*2} - 2c^*}{2c^* - 1} = 0.497$. This implies that for every value of $\bar{c}$ there is a value $c^*(\bar{c})$ which is an increasing function of $\bar{c}$. For $\underline{c} > c^*$, the only relevant threshold is that of comparing the offers $t = \{0, \bar{c}\}$ and $t = \{0, \underline{c}\}$, p_{23} , and this completes the proof.

B Additional numerical examples

In this Appendix, we provide additional numerical examples to illustrate the robustness of our main result to richer type distributions. In the baseline analysis, we assume two cost types in order to keep the exposition simple. In Table 5, we allow for three distinct cost levels and compute, for a grid of candidate offer profiles (t_1, t_2) , the induced success probabilities under unanimity and majority, the associated transfers, and the proposer's expected payoffs. The example below shows that, even when acceptance costs are more finely distributed, payoff-maximizing behavior can still generate a higher implementation probability under unanimity than under majority, thereby confirming the unanimity-majority paradox beyond the two-type benchmark. We also show that the mechanism behind the paradox is not tied to having only two responders; see Table 6.

Table 5 The responders' acceptance costs take values in $\{0, 0.08, 0.22\}$ with probabilities $\{0.5, 0.2, 0.3\}$, so that p_i is the cumulative probability $\Pr(c \leq t_i)$. For each offer (t_1, t_2) the table reports the success probability under unanimity (U) and majority (M), the total transfers $t_1 + t_2$ if the reform is implemented, and the proposer's expected payoff. We highlight the equilibrium payoffs and probabilities of success in bold. Under unanimity, the payoff-maximizing offer is $(0.22, 0.22)$, which guarantees acceptance (Success U = 1) and yields a payoff of 0.56. Under majority, the best offer is $(0, 0.08)$, which maximizes the proposer's payoff (0.782) but succeeds with probability only 0.85. Thus, in equilibrium, the unanimity rule delivers a strictly higher probability of reform acceptance than the majority rule, even though unanimity has a stricter approval requirement

Offers (t_1, t_2)	(p_1, p_2)	Success U	Success M	Transfers	Payoff U	Payoff M
(0, 0)	(0.5, 0.5)	0.25	0.75	0.000	0.250	0.750
(0, 0.08)	(0.5, 0.7)	0.35	0.85	0.080	0.322	0.782
(0, 0.22)	(0.5, 1.0)	0.50	1.00	0.220	0.390	0.780
(0.08, 0.08)	(0.7, 0.7)	0.49	0.91	0.160	0.412	0.764
(0.08, 0.22)	(0.7, 1.0)	0.70	1.00	0.300	0.490	0.700
(0.22, 0.22)	(1.0, 1.0)	1.00	1.00	0.440	0.560	0.560

Table 6 Three responders with i.i.d. costs $c_i \in \{0.05, 0.25\}$ and $\Pr(c_i = 0.05) = 0.55$. Here $p_i = \Pr(c_i \leq t_i)$ is the acceptance probability of responder i . “Success U/M” are the implementation probabilities under unanimity/majority. We highlight the equilibrium payoffs and probabilities of success in bold. The unanimity-optimal offer is (0.25, 0.25, 0.25); the majority-optimal offer (up to permutation) is (0.05, 0.05, 0.25)

Offers (t_1, t_2, t_3)	(p_1, p_2, p_3)	Success U	Success M	Transfers	Payoff U	Payoff M
(0, 0.05, 0.05)	(0, 0.55, 0.55)	0.000	0.303	0.100	0.000	0.272
(0, 0.05, 0.25)	(0, 0.55, 1)	0.000	0.550	0.300	0.000	0.385
(0, 0.25, 0.25)	(0, 1, 1)	0.000	1.000	0.500	0.000	0.500
(0.05, 0.05, 0.05)	(0.55, 0.55, 0.55)	0.166	0.575	0.150	0.141	0.489
(0.05, 0.05, 0.25)	(0.55, 0.55, 1)	0.303	0.798	0.350	0.197	0.518
(0.05, 0.25, 0.25)	(0.55, 1, 1)	0.550	1.000	0.550	0.247	0.450
(0.25, 0.25, 0.25)	(1, 1, 1)	1.000	1.000	0.750	0.250	0.250

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References

- Abreu, Dilip, and Faruk Gul. 2000. Bargaining and reputation. *Econometrica* 68 (1): 85–117. <https://doi.org/10.1111/1468-0262.00094>.
- Agranov, Marina. 2022. *Legislative bargaining experiments*. In *Bargaining: Current research and future directions*, edited by Emin Karagözoğlu and Kyle B. Hyndman, 179–202. Cham: Springer International Publishing. https://doi.org/10.1007/978-3-030-76666-5_9.
- Agranov, Marina, Hülya. Eraslan, and Chloe Tergiman. 2024. Bargaining in the shadow of uncertainty. *American Economic Journal: Microeconomics* 16 (4): 229–58. <https://doi.org/10.1257/mic.20220327>.
- Austen-Smith, David, and Timothy J. Feddersen. 2006. Deliberation, preference uncertainty, and voting rules. *American Political Science Review* 100 (2): 209–217. <https://doi.org/10.1017/S0003055406062113>.
- Azrieli, Yaron. 2018. The price of ‘one person, one vote’. *Social Choice and Welfare* 50 (2): 353–385. <https://doi.org/10.1007/s00355-017-1087-z>.
- Bailer, Stefanie, Mikko Mattila, and Gerald Schneider. 2015. Money makes the EU go round: The objective foundations of conflict in the council of ministers. *Journal of Common Market Studies* 53 (3): 437–456. <https://doi.org/10.1111/jcms.12194>.
- Baron, David P., and John A. Ferejohn. 1989. Bargaining in legislatures. *American Political Science Review* 83 (4): 1181–1206. <https://doi.org/10.2307/1961664>.

- Bellenghi, Guido. EU 2.0 revisited: Between vetocracy and rule of law concerns. Technical report.. (2022). <https://www.europeanlawblog.eu/pub/eu-2-0-revisited-between-vetocracy-and-rule-of-law-concerns/release/1>. Accessed July 20, 2025.
- Bouton, Laurent, Aniol Llorente-Saguer, and Frédéric. Malherbe. 2017. Unanimous rules in the laboratory. *Games and Economic Behavior* 102:179–198. <https://doi.org/10.1016/j.geb.2016.12.001>.
- Bouton, Laurent, Aniol Llorente-Saguer, and Frédéric. Malherbe. 2018. Get rid of unanimity rule: The superiority of majority rules with veto power. *Journal of Political Economy* 126 (1): 107–149. <https://doi.org/10.1086/695477>.
- Buchanan, James M. 1975. Public finance and public choice. *National Tax Journal* 28 (4): 383–394. <https://doi.org/10.1086/NTJ41863132>.
- Buchanan, James M., and Gordon Tullock. 1962. *The calculus of consent: Logical foundations of constitutional democracy*. Ann Arbor, MI: University of Michigan Press.
- Cecchini, Paolo. 1988. *The European challenge, 1992: The benefits of a single market*. Commission of the European Communities: Technical report.
- Chen, Ying, and Hülya. Eraslan. 2013. Informational loss in bundled bargaining. *Journal of Theoretical Politics* 25 (3): 338–362. <https://doi.org/10.1177/0951629813482232>.
- Chen, Ying, and Hülya. Eraslan. 2014. Rhetoric in legislative bargaining with asymmetric information. *Theoretical Economics* 9 (2): 483–513. <https://doi.org/10.3982/TE821>.
- Draghi, Mario. The future of European competitiveness. Technical report. Brussels: European Commission. (2024) https://commission.europa.eu/document/download/97e481fd-2dc3-412d-be4c-f152a8232961_en.
- Feddersen, Timothy, and Wolfgang Pesendorfer. 1998. Convicting the innocent: The inferiority of unanimous jury verdicts under strategic voting. *American Political Science Review* 92 (1): 23–35. <https://doi.org/10.2307/2585926>.
- Gonzalez-Sanchez, Eric, and Gino Loyola. 2024. Ultimatum bargaining with envy under incomplete information. *Mathematical Social Sciences* 127:1–11. <https://doi.org/10.1016/j.mathsocsci.2023.11.001>.
- Guarnaschelli, Serena, Richard D. McKelvey, and Thomas R. Palfrey. 2000. An experimental study of jury decision rules. *American Political Science Review* 94 (2): 407–423. <https://doi.org/10.2307/2586020>.
- Howarth, David. 1994. The compromise on Denmark and the treaty on European Union: A legal and political analysis. *Common Market Law Review* 31 (4): 765–805. <https://doi.org/10.54648/cola1994039>.
- Johnson, Marianne. 2006. The Wicksellian unanimity rule: The competing interpretations of Buchanan and Musgrave. *Journal of the History of Economic Thought* 28 (1): 57–79. <https://doi.org/10.1080/10427710500509755>.
- Konrad, Kai A., and Marcel Thum. 2021. The better route to global tax coordination: Gradualism or multi-lateralism? *Canadian Journal of Economics/Revue canadienne d'économie* 54 (2): 793–811. <https://doi.org/10.1111/caje.12513>.
- Laruelle, Annick, and Mika Widgrén. 1998. Is the allocation of voting power among EU states fair? *Public Choice* 94 (3): 317–339.
- Leech, Dennis. 1988. The relationship between shareholding concentration and shareholder voting power in British companies: A study of the application of power indices for simple games. *Management Science* 34 (4): 509–527.
- Levy, Richard E. 2017. The law and economics of supranationalism: the European Union and the subsidiarity principle in collective action perspective. *European Journal of Law and Economics* 43:441–473. <https://doi.org/10.1007/s10657-015-9508-x>.
- Ma, Zizhen. 2023. Efficiency and surplus distribution in majoritarian reputational bargaining. *Journal of Economic Theory* 210 : 105649. <https://doi.org/10.1016/j.jet.2023.105649>.
- Mattila, Mikko, and Jan-Erik. Lane. 2001. Why unanimity in the council? A roll call analysis of council voting. *European Union Politics* 2 (1): 31–52. <https://doi.org/10.1177/1465116501002001002>.
- McGann, Anthony J. 2004. The tyranny of the supermajority: How majority rule protects minorities. *Journal of Theoretical Politics* 16 (1): 53–77. <https://doi.org/10.1177/0951629804038902>.
- Miller, Luis, and Christoph Vanberg. 2013. Decision costs in legislative bargaining: An experimental analysis. *Public Choice* 155:373–394. <https://doi.org/10.1007/s11127-011-9866-z>.
- Miller, Luis, and Christoph Vanberg. 2015. Group size and decision rules in legislative bargaining. *European Journal of Political Economy* 37:288–302. <https://doi.org/10.1016/j.ejpoleco.2014.09.005>.
- Moravcsik, Andrew. 1998. *The choice for Europe: Social purpose and state power from Messina to Maastricht*. Ithaca, NY: Cornell University Press. <https://doi.org/10.2307/2586098>.
- Mueller, Dennis C. 2003. *Public Choice III*. Cambridge: Cambridge University Press.
- Musgrave, Richard Abel. 1939. The voluntary exchange theory of public economy. *The Quarterly Journal of Economics* 53 (2): 213–237. <https://doi.org/10.2307/1882886>.
- Palfrey, Thomas R. “Experiments in political economy.” In Handbook of experimental economics, edited by John H. Kagel and Alvin E. Roth, 2:347–434. Princeton, NJ: Princeton University Press. (2016)

- Piazzolo, David, and Christoph Vanberg. 2025. Legislative bargaining with private information: A comparison of majority and unanimity rule. *Games and Economic Behavior* 153:499–522. <https://doi.org/10.1016/j.geb.2025.07.010>.
- Rosenthal, Howard, and William R. Zame. 2022. Sequential referenda with sophisticated voters. *Journal of Public Economics* 212 : 104702. <https://doi.org/10.1016/j.jpubeco.2022.104702>.
- Schuette, Leonard. Should the EU make foreign policy decisions by majority voting? Technical report. London: Center for European Reform. (2019) <https://cer.eu/publications/archive/policy-brief/2019/should-eu-make-foreign-policy-decisions-majority-voting>.
- Svensson, Palle. 1994. The Danish yes to Maastricht and Edinburgh: The EC referendum of May 1993. *Scandinavian Political Studies* 17 (1): 69–82. <https://doi.org/10.1111/j.1467-9477.1994.tb00050.x>.
- Tsai, Tsung-Sheng., and C. C. Yang. 2010. On majoritarian bargaining with incomplete information. *International Economic Review* 51 (4): 959–979. <https://doi.org/10.1111/j.1468-2354.2010.00607.x>.
- Tsebelis, George. 2002. *Veto players: How political institutions work*. Princeton, NJ: Princeton University Press.
- Wicksell, Knut. 1896. *Finanztheoretische untersuchungen*. Jena, Germany: Verlag von Gustav Fischer.
- Wicksell, Knut. 1958. A new principle of just taxation. In *Classics in the theory of public finance*, ed. Richard A. Musgrave and Alan T. Peacock, 72–118. London: Palgrave Macmillan.

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